

Data Source Examination – Port Authority Capstone Project

Technical Report

DATA SYNTHESIS & INTEGRATION PHASE

Data Source Examination & Structural Analysis

1 Traffic Dataset — Detailed Structural Examination

Dataset Overview

Metric	Value
Total Rows:	5,383,378
Total Columns:	30
Memory Usage:	~1.2 GB
Duplicate Rows:	0

This is the largest dataset among all available sources and represents the core operational traffic dataset. The dataset contains detailed hourly, lane-level traffic activity across multiple facilities.

Time Coverage

From inspection:

- Earliest visible date: 2022-09-15
- Latest visible date: 2025-05-22
- Year column ranges: 2022 – 2025

This indicates approximately 3 years of continuous operational data.

Granularity

The dataset operates at: **Hourly × Lane × Facility level**

Columns such as DATE, TIME, LANE, FAC, FAC_B, and FAC_G confirm highly detailed operational granularity. This dataset is not aggregated — it is transactional traffic flow data.

Core Metrics

Key measurable variables include:

- TOTAL traffic
- CLASS 1 to CLASS 11 vehicle classifications
- CASH, EZPASS, VIOLATION
- Aggregated categories: Autos, Small_T, Large_T, Buses
- Facility grouping: FAC_G, FAC_G2
- LANEMODE
- Day_Name

Missing Values Analysis

Core operational columns contain zero missing values:

- DATE, FAC, LANE, TIME
- TOTAL, CASH, EZPASS, VIOLATION
- Autos, Small_T, Large_T, Buses, Yr

However, significant missing values exist in vehicle class breakdown columns:

- CLASS 2 → 407,007 missing
- CLASS 6 → 1,354,198 missing
- CLASS 11 → 1,011,351 missing

This suggests that vehicle classification is not consistently captured across all facilities or lanes.

Descriptive Statistics Highlights

- Mean TOTAL traffic per hour: 270.17
- Median TOTAL traffic: 199
- Maximum hourly TOTAL traffic: 3,938
- TIME ranges from 0 to 2300, confirming hourly format
- Yr ranges from 2022 to 2025

Traffic distribution is highly skewed, indicating peak-hour concentration.

Facility Coverage

- FAC_B unique facilities: 8
- FAC_G facility groups: 3
- FAC_G2 sub-groups: 6
- LANEMODE unique values: 3
- Day_Name: 7 unique values

The dataset covers multiple bridges and tunnels under Port Authority operations.

Key Observations

- Extremely large and operationally detailed dataset
- No duplicate rows
- Core metrics are complete
- Vehicle class breakdown partially missing
- Highly granular structure suitable for aggregation

This dataset serves as the primary backbone dataset for integration.

2 Bus Dataset — Detailed Structural Examination

Dataset Overview

Metric	Value
Total Rows:	2,784
Total Columns:	4
Memory Usage:	~87 KB
Duplicate Rows:	0

This dataset is significantly smaller and fully aggregated.

Time Coverage

- Earliest visible Start_Date: 2021-02-08
- Latest visible End_Date: 2025-01-31
- Total unique Start_Date values: 232

Each row represents a 5-day period (Monday–Friday). This dataset spans approximately 4 years.

Granularity

The dataset operates at: **Weekly × Carrier level**

Columns: Start_Date, End_Date, Carrier, Volume. Each row represents weekly bus movement totals per carrier.

Missing Values

No missing values in any column.

Descriptive Statistics

Volume statistics:

- Mean: 221.71
- Median: 29
- Maximum: 2,577
- Standard Deviation: 614.33

The distribution is highly skewed. This indicates a small number of dominant carriers (e.g., NJ Transit).

Carrier Coverage

- Unique carriers: 13
- The dataset contains 232 unique weekly start dates, indicating consistent weekly reporting across carriers.

This indicates stable weekly reporting over multiple years.

Key Observations

- Clean and structured dataset
- No duplicates
- Weekly aggregation
- Carrier-based structure
- Highly skewed distribution

This dataset represents weekly bus movement counts at the Port Authority Bus Terminal.

3 Passenger Dataset — Detailed Structural Examination

Dataset Overview

Metric	Value
Total Rows:	2,786
Total Columns:	4
Memory Usage:	~87 KB
Duplicate Rows:	1

Structurally similar to Bus dataset.

Time Coverage

- Earliest visible Start_Date: 2020-12-07
- Latest visible End_Date: 2025-01-31
- Unique Start_Date values: 232

This dataset has slightly longer historical coverage than the Bus dataset.

Granularity

Operates at: **Weekly × Carrier level**. Same structure as Bus dataset.

Missing Values

- Start_Date → 2 missing
- End_Date → 2 missing
- Carrier → 2 missing
- Volume → 2 missing

Unlike Bus dataset, this dataset contains minor inconsistencies.

Descriptive Statistics

Volume statistics:

- Mean: 6,177.60
- Median: 796
- Maximum: 90,225
- Standard Deviation: 18,033.24

Extremely skewed distribution. Passenger volumes are significantly larger than bus volumes.

Carrier Coverage

- Unique carriers: 15

Passenger dataset includes 2 additional carriers not present in Bus dataset.

Key Observations

- Minor data quality issues (missing + 1 duplicate)
- Weekly aggregation
- Carrier-based
- Highly skewed passenger distribution
- Largest passenger week recorded: 90,225

This dataset represents weekly passenger volumes per carrier at PABT.

4 Speeds Dataset — Detailed Structural Examination

Dataset Overview

Metric	Value
Total Rows:	192
Total Columns:	7
Memory Usage:	~10 KB
Duplicate Rows:	0

Small, aggregated performance dataset.

Time Coverage

- Unique Month_Year values: 17
- Covers approximately 17 months
- Latest visible Month_Year: 2025-04-01

This dataset has shorter historical coverage than other datasets.

Granularity

Operates at: **Monthly × Facility × Direction level**

Columns: Facility, Freeflow, Month_Year, Avg_Speed, Direction, Delta, Facility_Order.

Descriptive Statistics

- **Freeflow:** Mean: 42.02, Min: 32, Max: 47.9
- **Avg_Speed:** Mean: 35.73, Min: 13, Max: 49.5
- **Delta:** Mean: -6.52, Min: -27.3, Max: 4.1

Delta being negative indicates that average speeds are generally below freeflow conditions.

Facility Coverage

- Unique Facilities: 6
- Directions: 2 (EB, WB)

Not all traffic facilities appear in speeds dataset.

Key Observations

- Clean dataset
- Monthly aggregation
- Facility-level
- Shorter historical span
- Captures congestion performance via Delta

Cross-Dataset Structural Comparison

Dataset	Rows	Granularity	Time Span	Issues
Traffic	5,383,378	Hourly × Lane	2022–2025	Missing in class columns
Bus	2,784	Weekly × Carrier	2021–2025	None
Passenger	2,786	Weekly × Carrier	2020–2025	2 missing rows, 1 duplicate
Speeds	192	Monthly × Facility	~17 months	None

Overall Structural Understanding

The datasets operate at three different time granularities:

- Hourly (Traffic)
- Weekly (Bus & Passenger)
- Monthly (Speeds)

They also represent different entity dimensions:

- Facility-based (Traffic, Speeds)
- Carrier-based (Bus, Passenger)

Conclusion: This multi-dimensional structure requires careful time alignment and aggregation for integration.